SURVEY · 495 PAPERS · 2020–2026

Computationally Invisible

Charting the NLP Landscape of Iranic Languages

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SilkRoadNLP submission · companion website: sigiranic.github.io



For most of the other varieties, no paper, benchmark or tool was found.

Slides drawn in Excalidraw; every number is from our paper (see references/digest.md)

Our survey in one slide

Question

Where does NLP stand for the Iranic language family?

- 101 ISO 639-3 coded varieties
- over 200 million speakers
- from the Caucasus to Xinjiang, from Russia to Oman
- four living branches

What we did

A systematic survey:

- 5,878 candidate papers, 2020 – May 2026
- LLM filter + annotation (Qwen3.5-9B), checked by two of us
- 495 papers kept
- audit of corpora, Wikipedia, morphology, LID, parallel data

What we found

Severe imbalance:

- Persian: 394 papers (80%)
- Pashto: 51M speakers, 29 papers
- Balochi: 9M speakers, 4 papers
- most varieties: nothing found

Persian itself is saturating on standard tasks; the rest need their first benchmark.

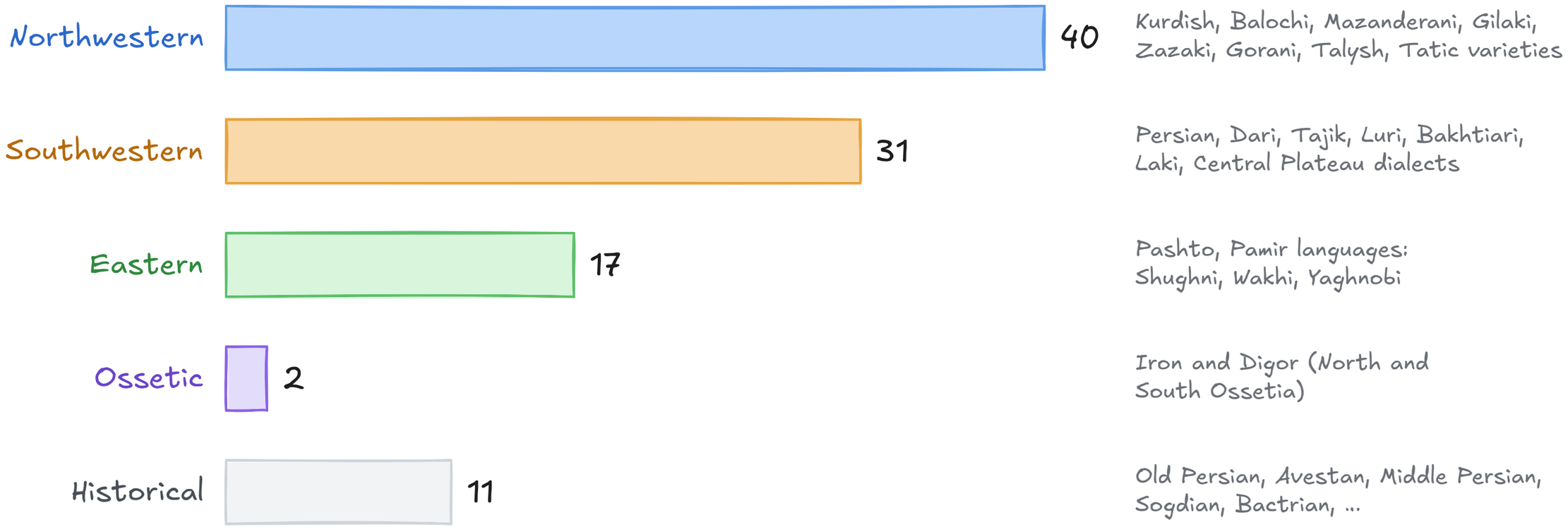
Iranic NLP is mostly Persian NLP; the other varieties are computationally invisible.

Glossary

term	meaning
ISO 639-3	international three-letter language codes, e.g. pes = Iranian Persian, ckb = Sorani
macrolanguage	one ISO code that covers several individual codes, e.g. fas covers pes and prs
variety	neutral word for a language or dialect; avoids deciding which one it is
FineWeb2	large multilingual web-text corpus; a proxy for how much text a language has online
Wiktionary · UniMorph	community dictionary · database of inflected word forms (paradigms)
LID	language identification: which language is this text? Needed to build corpora
parallel data	the same sentences in two languages; the raw material of machine translation
FLORES+	standard translation benchmark, about 2K sentences per language
pivot language	translating through a third language (here Persian or English)
ZWNJ	zero-width non-joiner: an invisible character that splits Persian compound words
taarof	Persian ritual politeness (e.g. refusing an offer before accepting it)

101 coded varieties in four living branches

Counted from the appendix table of all 101 ISO 639-3 codes



Types: 50 dialects · 36 languages · 11 historical · 4 macrolanguages (fas, kur, pus, bal)

Most codes are small Northwestern and Southwestern varieties, many of them dialects.

The major modern languages

L1 speakers, script and NLP resource level (Table 1)

Southwestern

Persian	91M	Perso-Arabic, Cyrillic	high
Dari	15M	Perso-Arabic	medium
Tajik	8M	Cyrillic	low
Luri / Bakhtiari	5M	Perso-Arabic	very low
Hazaragi	3M	Perso-Arabic	very low
Laki	0.7M	Perso-Arabic	very low

Eastern

Pashto	51M	Perso-Arabic	low
Shughni	0.1M	Cyrillic, Latin	very low
Wakhi	0.06M	Perso-Ar., Cyr., Latin	very low
Yaghnobi	0.01M	Cyrillic	very low

Northwestern

Kurdish	26M	Perso-Ar., Latin, Cyr.	low
Balochi	9M	Perso-Arabic	very low
Gilaki	3M	Perso-Arabic	very low
S. Kurdish	3M	Perso-Arabic	very low
Zazaki	2–4M	Latin	very low
Mazanderani	1.4M	Perso-Arabic	very low
Talysh	0.8M	Perso-Ar., Cyr., Latin	very low
Gorani	0.4M	Perso-Arabic	very low

Ossetic

Ossetic	0.6M	Cyrillic	very low

NLP level is our rating of available resources. One language is rated high, one medium, three low and the rest very low.

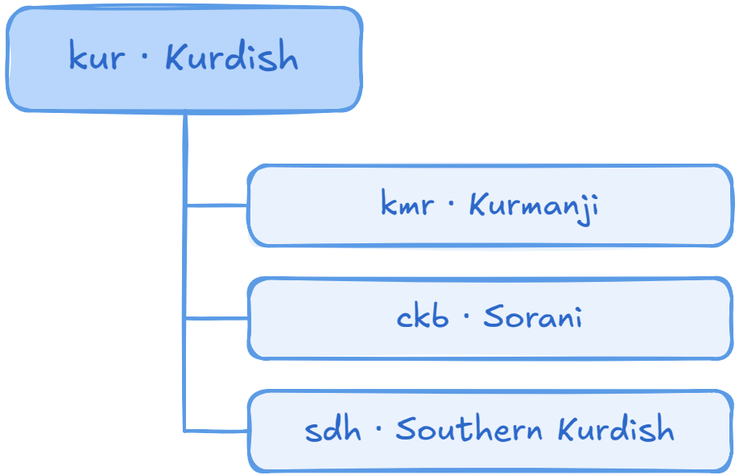
Computationally Invisible (SilkRoadNLP submission), Table 1. Speaker numbers are approximations from several sources

What counts as one language?

The ISO labels do not match how the varieties are used



Largely mutually intelligible; whether they are one language is politically charged.



Limited mutual intelligibility, Latin and Arabic-based scripts: an unqualified 'Kurdish' in a paper is ambiguous.

Language or dialect?

Laki: a Kurdish dialect to some, a separate language to others. Bakhtiari: its own code, yet Persian speakers sometimes treat all Luri as Persian. These choices decide what data is labelled and who gets tools.

A model trained on 'fas' or 'Kurdish' may not serve the people the label suggests.

From 5,878 candidates to 495 papers



Query: every Iranic language name and ISO code × NLP task terms (sentiment, QA, summarization, benchmark, corpus, …), plus targeted searches and citation tracing.

The six annotated fields

1 language(s) 2 NLP task 3 resource type
4 public availability 5 year 6 venue

then deduplication; multilingual papers count once per language

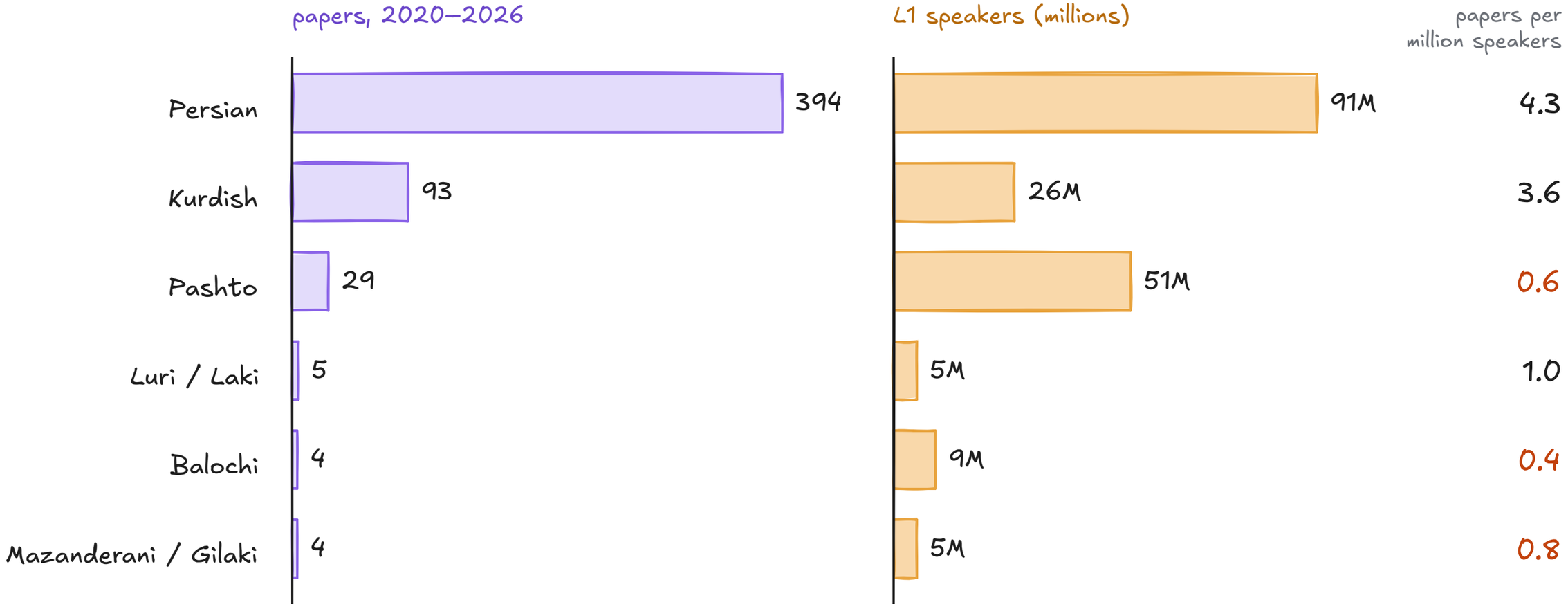
Quality control and caveat

- every annotation checked by two of us; a third author breaks ties
- prompts and code are public
- 'no resource exists' means: none publicly documented was found in this search

An LLM does the first pass; humans check every label.

Papers do not follow speakers

Paper count and L1 speakers for the major languages



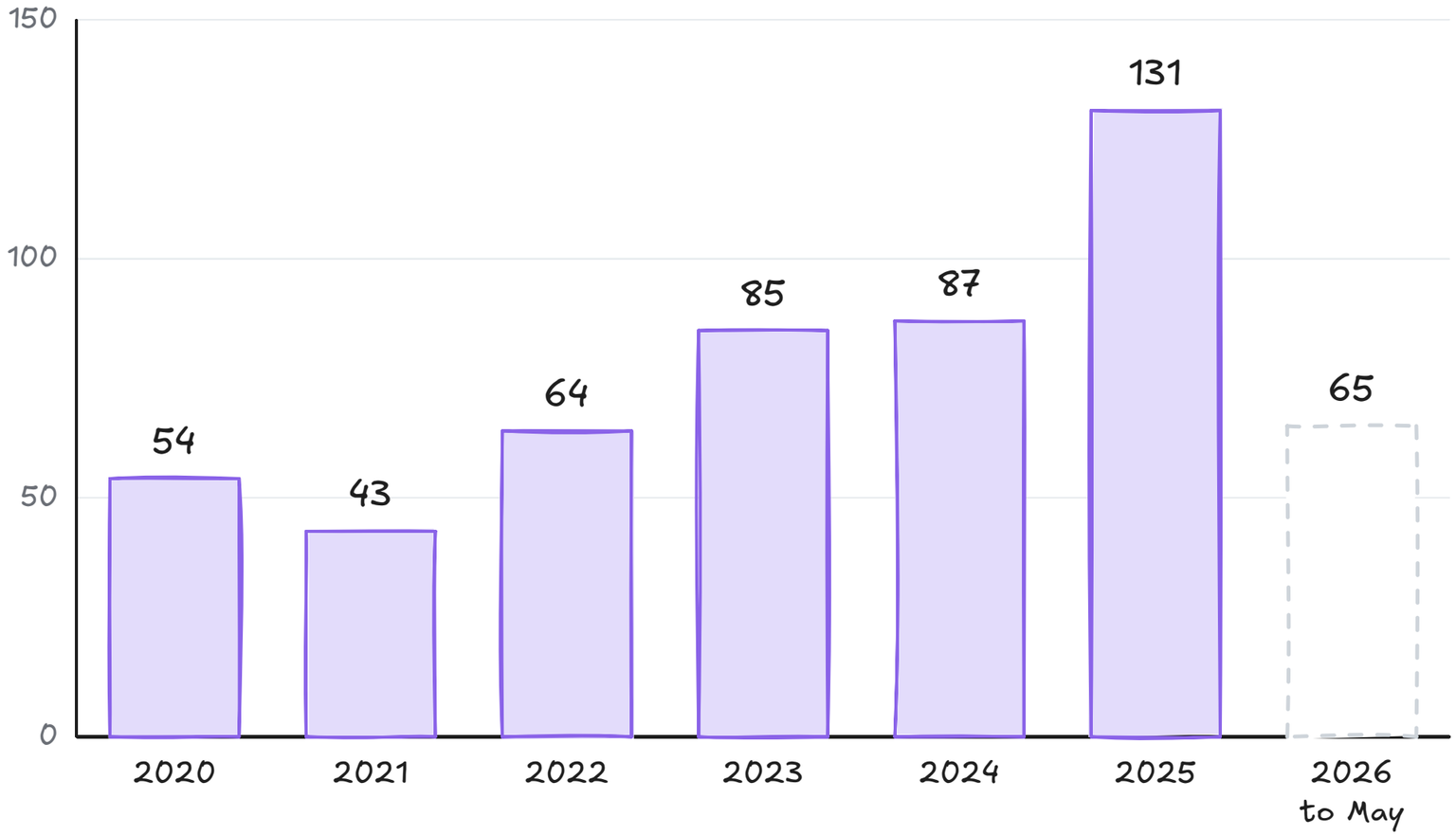
Persian: 80% of the papers, about 60% of the family's speakers. Red: fewer than one paper per million speakers.

Pashto has twice Kurdish's speakers but a third of its papers.

Computationally Invisible (SilkRoadNLP submission), §3 and Figure 2; speakers from Table 1. Multilingual papers count once per language

The field is growing, but mostly in Persian

Surveyed papers per year



Reading the bars

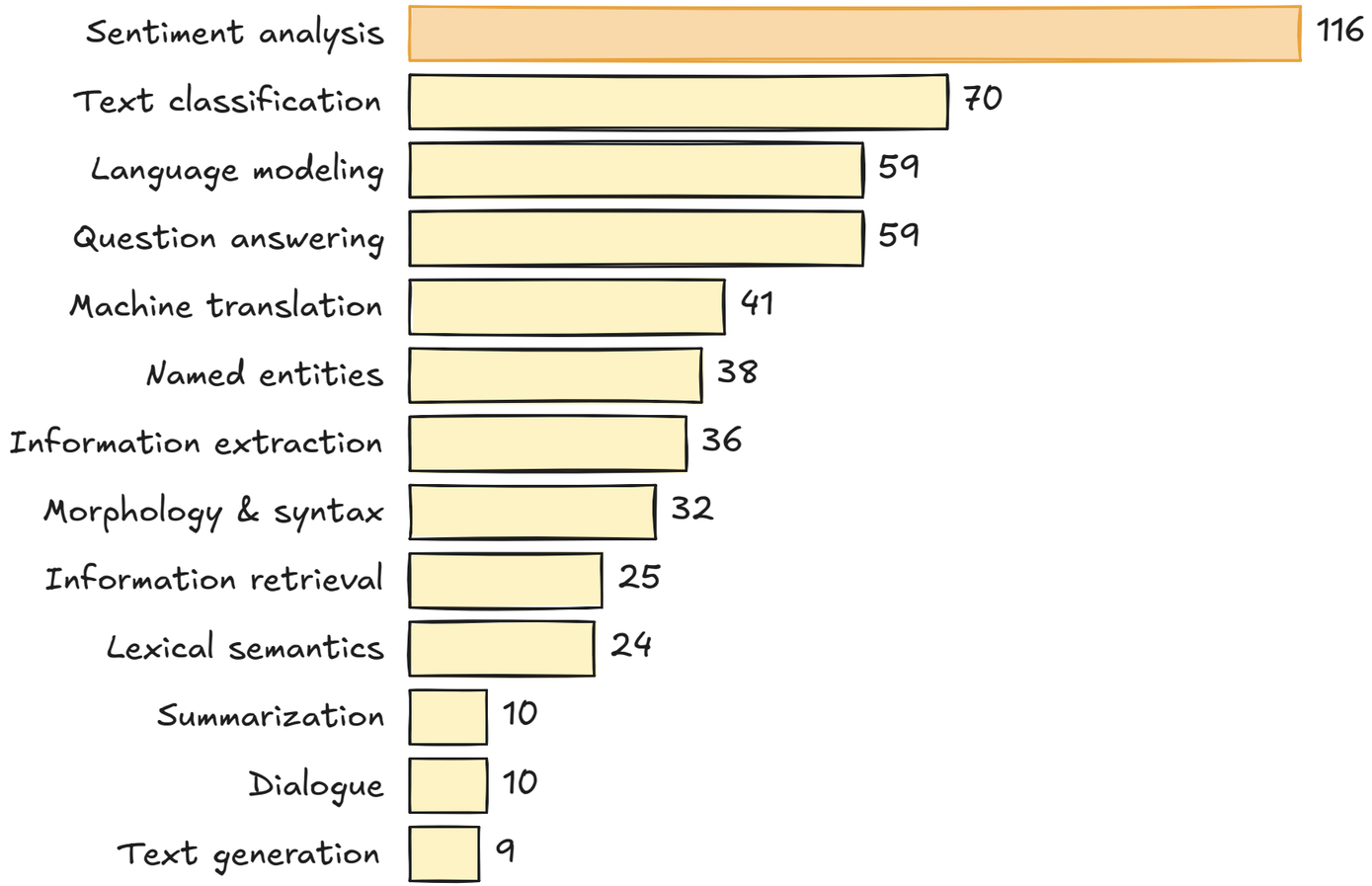
- 54 papers in 2020, 131 in 2025
- growth is driven almost entirely by Persian
- Kurdish and Pashto grow modestly; the other varieties stay near zero
- 2026 is partial: the search ended in May

More papers do not mean more languages.

Computationally Invisible (SilkRoadNLP submission), §3 and Figure 3 (values read from the figure)

Which tasks get studied

Papers per NLP task; a paper with several tasks counts once per task



What stands out

- classification-style tasks dominate: sentiment alone has 116 papers
- generation and dialogue have about 10 each
- most attention goes to Persian in every task
- for non-Persian varieties, machine translation is the most active area

Classification dominates; generation and dialogue are barely studied.

The digital footprint behind the papers

Selected rows of our resource table (Table 2), with Arabic and Turkish for reference

language	FineWeb2 web text	Wikipedia articles	Wiktionary entries	UniMorph forms	grammar · WordNet spell · ASR
Arabic (ref.)	106 GB	1,293K	74,093	140,003	✓ ✓ ✓ ✓
Turkish (ref.)	135 GB	658K	43,253	275,460	✓ ✓ ✓ ✓
Persian	102 GB	1,075K	19,248	37,128	✓ ✓ ✓ ✓
N. Kurdish	979 MB	92K	8,254	216,370	✓ ✓ ✓ ✓
Tajik	—	118K	430	0	✗ ✗ ✗ ✗
Pashto	—	21K	1,100	6,945	✗ ✗ ✗ ✗
Gilaki	16.4 MB	48K	34	0	✓ ✗ ✗ ✗
Mazanderani	1.7 MB	65K	337	0	✗ ✗ ✗ ✗
Balochi	—	1.2K	429	0	✗ ✗ ✗ ✗
Laki	57.7 KB	Incubator	85	0	✓ ✗ ✗ ✗
N. Luri	82.7 KB	—	5	0	✓ ✗ ✗ ✗

— = no data. Red = absent or tiny. Tools: ✓ available, ✗ absent.

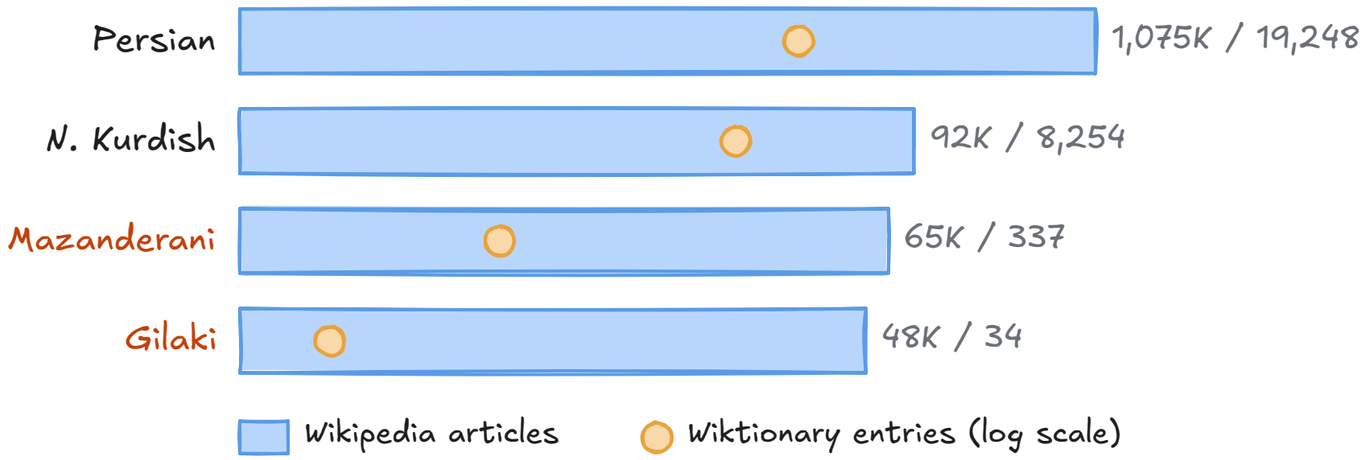
Below Persian and Kurmanji, the columns thin out fast.

Wikipedia: present, absent, or hollow

The 97 individual varieties (May 2026)



Hollow presence: many articles, almost no dictionary



A failed Wikipedia

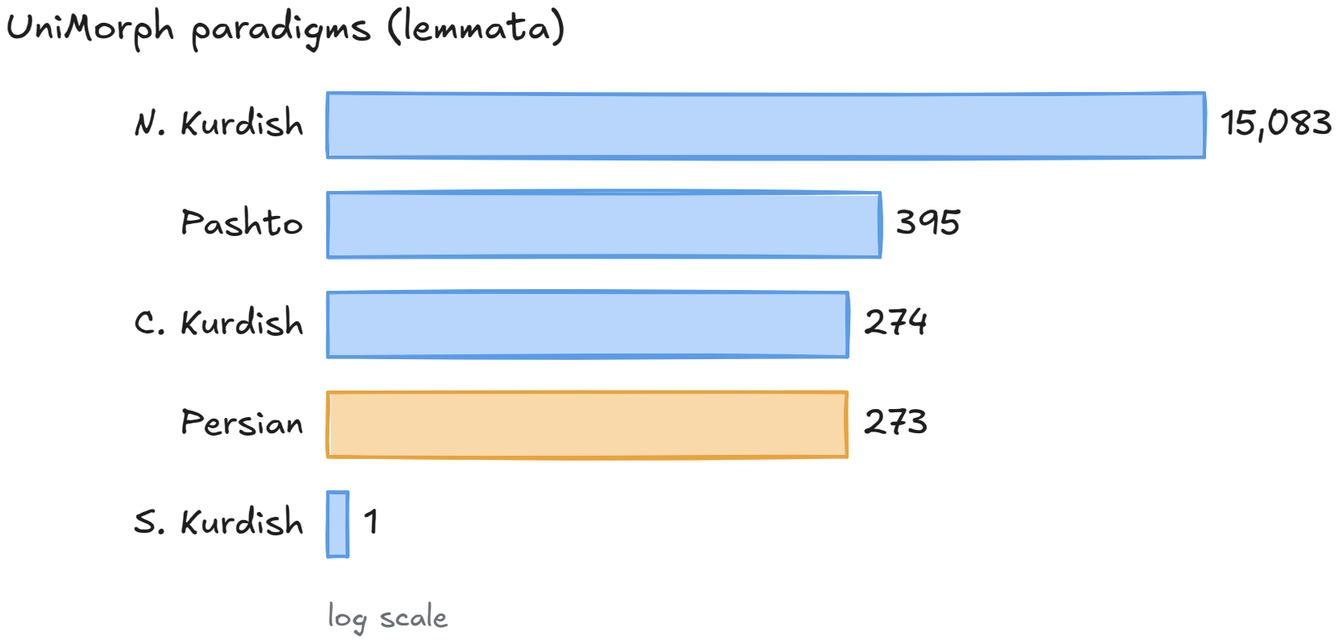
The Northern Luri Wikipedia was abandoned: reviewers found Persian with light word substitution, not native Luri grammar.

Article counts can hide how little real language data there is.

Morphology: rich languages, thin resources

Why morphology matters here

- Persian: Ezafe construction and rich verb inflection
- Kurdish: split ergativity
- Pashto: verbs agree with several arguments at once
- Pamir languages: up to five agreement slots per verb



Zero: Balochi, Talysh, Mazanderani, Gilaki, Gorani, Luri, Laki, Tajik, Zazaki

Tools: Persian has Hazm, DadmaTools and Parsivar; no equivalent exists for any other variety.
Treebanks: Persian UD, Kurmanji UD and Central Kurdish–Mukri only.

Persian has the text but only 273 paradigms; Kurmanji has 15,083.

Language identification: the first filter

Every web corpus is built by LID; confusable pairs make it hard

Hard to tell apart

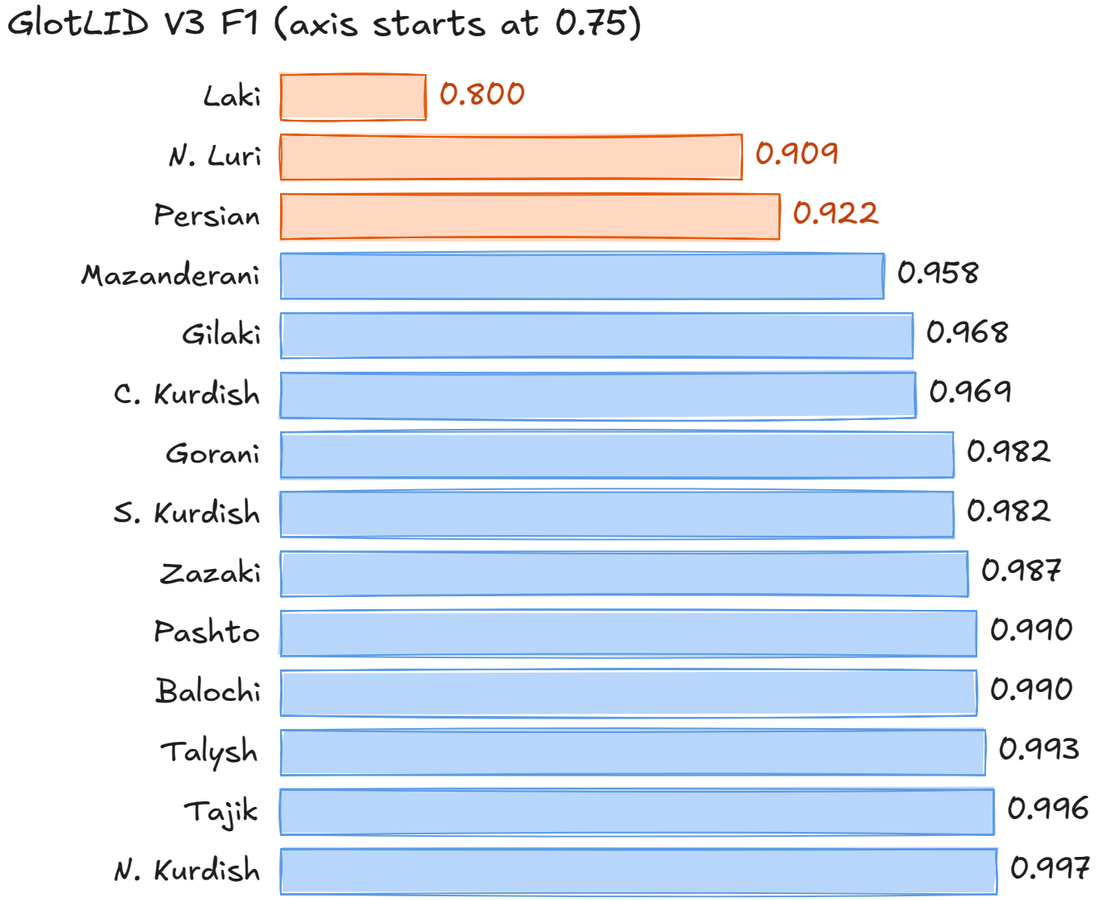
Persian ↔ Dari ↔ Tajik

Luri ↔ Persian

Talysh ↔ Azerbaijani

Shared vocabulary (Persian, Dari, Tajik) or heavy contact (Luri, Talysh).

GlotLID V3 covers 16 Iranic variety-script pairs; PALI is a LID benchmark for Perso-Arabic script with six Iranic varieties.

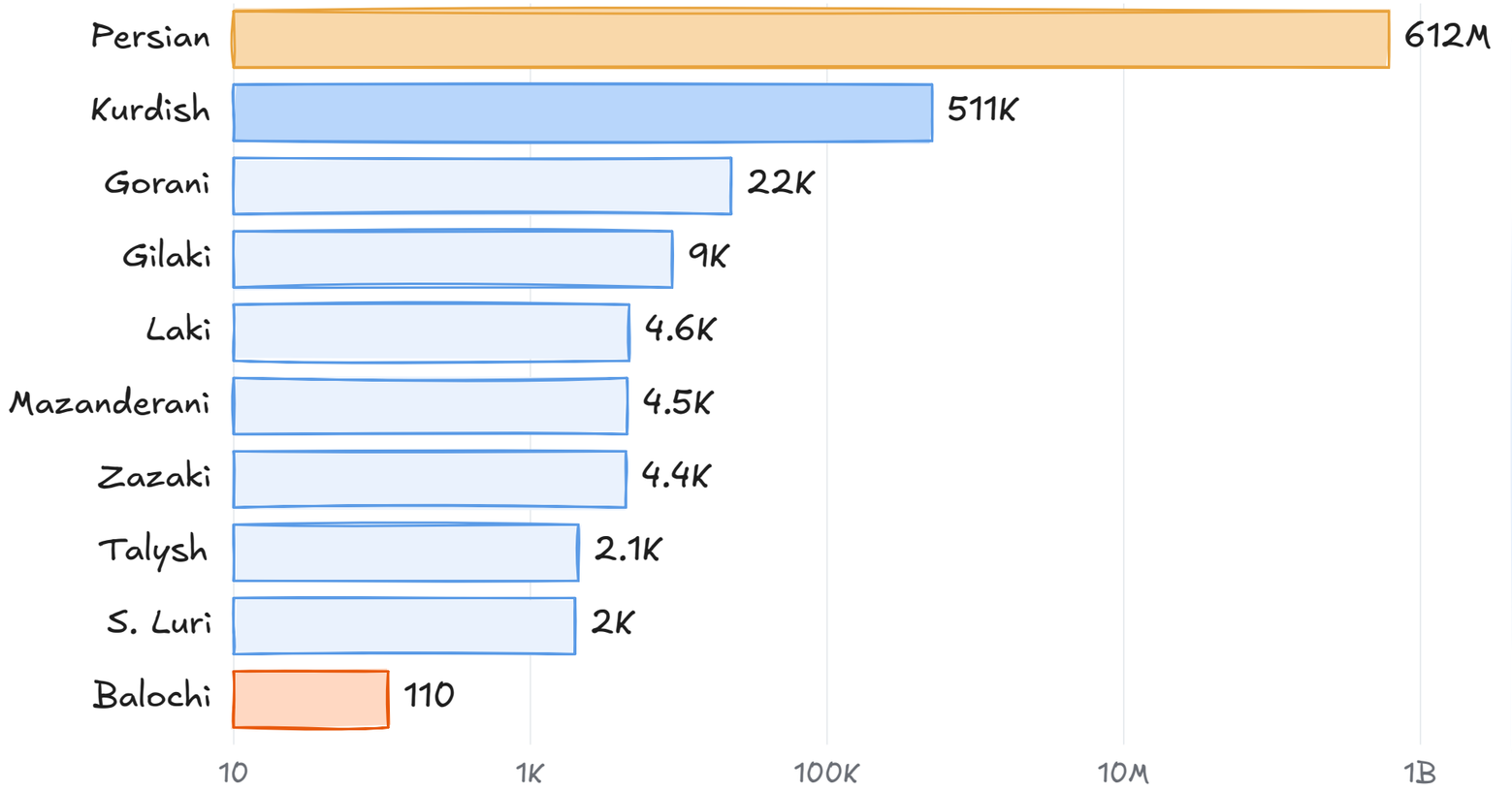


Even a high F1 can hide false positives that swamp a small language's corpus.

Computationally Invisible (SilkRoadNLP submission), §4 and Table 2 (Pashto and Balochi F1 reported for pbt and bcc)

Parallel data with English: three tiers

Sentence pairs with English, log scale



Three tiers

1 Persian: over 600M pairs (web crawls, subtitles, NLLB-200)

2 Kurdish: about 511K (Kurdish Parallel Corpus, plus SMOL and PARME)

3 everyone else: under 40K in total; Balochi: 110

Persian has over five million times as many English pairs as Balochi.

Between the smaller varieties: almost nothing

Parallel sentence pairs between Iranic varieties

	en	fas	kur	glk	mzn	tly	lki	luz	hac	diq
fas	612M									
kur	511K	326K								
glk	9K	5.5K	488							
mzn	4.5K	5.1K	1.6K	1.1K						
tly	2.1K	2.1K	1.6K	54	851					
lki	4.6K	3.4K	994	437	3K	849				
luz	2K	2K	994	39	1.4K	785	1.2K			
hac	22K	7.8K	1.6K	2K	1.3K	409	723	934		
diq	4.4K	4.4K	3.3K	0	0	0	0	0	0	
bal	110	30	35	0	0	0	0	0	0	0

fas = Persian, kur = Kurdish, glk = Gilaki, mzn = Mazanderani, tly = Talysh, lki = Laki, luz = S. Luri, hac = Gorani, diq = Zazaki, bal = Balochi

What the matrix shows

- below Persian and Kurdish, at most a few thousand pairs for any pair
- Zazaki: 0 pairs with all six smaller varieties
- Balochi: 30 with Persian, 35 with Kurdish, 0 otherwise
- Kurmanji–Sorani is the only substantial intra-Iranic MT corpus

Translation between neighbours has to go through Persian or English.

Computationally Invisible (SilkRoadNLP submission), Table 3 and §4

Task by task: Persian and everyone else

	Persian	other Iranic varieties
Pretraining	ParsBERT; Matina corpus (73B tokens) and LLM; FarsInstruct (21 instruction datasets)	Kurdish: KuBERT, KurdBERT; Pashto: Ps-BERT. Others: no monolingual model; XLM-R, Glot500
Benchmarks	ParsiNLU (14,500+ items, natively written); Khayyam (50+ domains); cultural benchmarks	Belebele (translated): Tajik, Sorani, S. Pashto. APARSIN: first beyond the top five varieties
Question answering	25+ datasets: PQuAD (80K), PersianMHQA (7K), PersianMedQA (20K)	one Sorani news dataset; no natively written QA for any other variety
Speech	DeepMine (480+ h ASR), ParsVoice (3,526 h TTS)	Sorani: JIRA, CORDI. DOLMA: 40+ h for six varieties (Zazaki 0.2 h); Whisper WER 37.9–89.5%
OCR, vision	VQA, captioning, printed/handwritten OCR	essentially absent
NER, classification, summarization	well studied	no annotated data

For most varieties, the first dataset of any kind is still to be built.

Persian: saturating and still open

Approaching saturation

- standard sentiment classification
- Wikipedia-based reading comprehension
- named entity recognition

Many benchmarks are translated from English (ParSQuAD from SQuAD).

Open frontiers

- cultural and pragmatic competence
- multi-hop reasoning
- code-switching
- robustness to dialects and colloquial Persian (HarfoSokhan: 6M colloquial→formal pairs)

Example: taarof (ritual politeness)

TaarofBench: LLMs score 40–48% below native Persian speakers on taarof scenarios. Frontier models do well on factual recall but show gaps in cultural knowledge and pragmatic judgment (Khayyam, FarsEval-PKBETS, ELAB). Taarof is absent from all existing evaluation frameworks.

Persian needs new kinds of tests; the other varieties need a first test.

Transfer from relatives: the main strategy

lexical proximity

Persian, Dari and Tajik share most vocabulary: strong zero-shot transfer, given correct script normalization

script proximity

all Perso-Arabic varieties share tokenization conventions, but Unicode differences silently degrade transfer

grammatical proximity

varieties in one branch share morphology, e.g. Kurmanji → Sorani MT; script normalization helps Kurdish

In practice

- cross-lingual transfer runs almost entirely through Persian or English as pivots
- a multilingual model that covers Persian does not cover Dari unless Dari data was included
- tokenizers are calibrated for high-resource languages and over-segment related varieties
- systematic intra-family transfer studies are still lacking

Relatedness is an asset, but nobody has measured it systematically yet.

Same letter, different bytes

Script heterogeneity: a concrete source of silent failures

yeh	ی	U+06CC Farsi yeh	≠	ي	U+064A Arabic yeh
kaf	ک	U+06A9 keheh	≠	ك	U+0643 Arabic kaf

More of the same

- ZWNJ inside Persian compounds
- right-to-left mixed with Latin text
- Kurmanji in Latin, Arabic or Cyrillic

In practice, ya and kaf code points differ across Persian, Pashto and Dari text, so string matching, deduplication and tokenization break without any error message. (Code points shown here as an illustration.)

Scripts in use

About a dozen scripts across the family, most only historical. Living varieties use Perso-Arabic, Latin and Cyrillic, often in parallel; Tajik uses Cyrillic only. Priority: one shared Iranic normalization library.

Normalize before you train, or related varieties look unrelated.

Six open challenges, one priority each

Resource concentration

Persian ≈ 80% of papers; Pashto (51M) has fewer resources than Kurdish

priority
shared tasks and central data hubs

Script heterogeneity

Perso-Arabic, Latin, Cyrillic; inconsistent code points; ZWNJ

priority
a unified Iranic normalization library

Dialectal fragmentation

most benchmarks test only the standard written form

priority
colloquial and regional benchmarks

Cultural misalignment

translated benchmarks; Persian sentiment, QA, NER near saturation

priority
benchmarks co-created with speakers

Endangered varieties

Pamir and Central Plateau varieties may disappear within decades

priority
multilingual speech models, community recording

Structural inequity

sanctions limit compute, API access and travel for many researchers

priority
compute and access programs

Three things the community can do now

1 · Raw text

Put Iranic text on the open web, in identifiable form.

Web crawls feed every corpus; untagged or mistagged text is invisible to LID-filtered pipelines.

2 · Translation benchmarks

Contribute to FLORES+ / OLDI.

FLORES+ grew from about 200 to 230 varieties, but still has the same 6 Iranic ones as FLORES-200: pes, prs, tgk, ckb, kmr, pbt.

3 · Evaluation beyond MT

At least one natively written benchmark per variety.

Shared tasks are the most plausible way to get them.

About 2K sentences per variety and an open pull-request process: the bottleneck is people, not infrastructure. The lesson from African NLP: community shared tasks and data hubs catalyse progress.

None of these needs a new model; they need contributors.

Contributions, scope and next steps

What this survey contributes

- the first systematic map of *NLP* for the whole family (101 codes)
- beyond paper counts: an audit of corpora, Wikipedia, morphology, LID and parallel data
- conservative wording: 'not found' rather than 'does not exist'
- concrete, low-cost priorities, plus a companion website and public code

Scope, and where we go next

- we searched indexed venues in English; Persian-language journals are a natural next addition to the live index
- *LLM*-assisted screening, checked by people; a recall audit of the filter is future work
- the *NLP* levels summarise resources at a glance; a scoring rule would make them reproducible
- paper counts show attention, not quality, and reflect sanctions and access limits

Takeaways

1 · One variety gets the attention

Iranian Persian: 394 of 495 papers (80%). Pashto (51M) has 29, Balochi (9M) has 4.

2 · The gap starts below the papers

Little or no web text, dictionary entries, UniMorph paradigms or tools for most varieties; under 40K parallel pairs.

3 · Labels and scripts are part of the problem

Macrolanguage codes, contested dialect status and inconsistent Unicode hide varieties from pipelines.

4 · Persian needs new tests, the rest need first tests

Persian is saturating on standard tasks; cultural and pragmatic competence is open.

5 · Small contributions go far

Tagged web text, FLORES+ sentences, one native benchmark per variety.

sigiranic.github.io · github.com/sigiranic/sigiranic.github.io